



Nothin' but a G Thang

The Evolution of Cellular Networks

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Who am I?



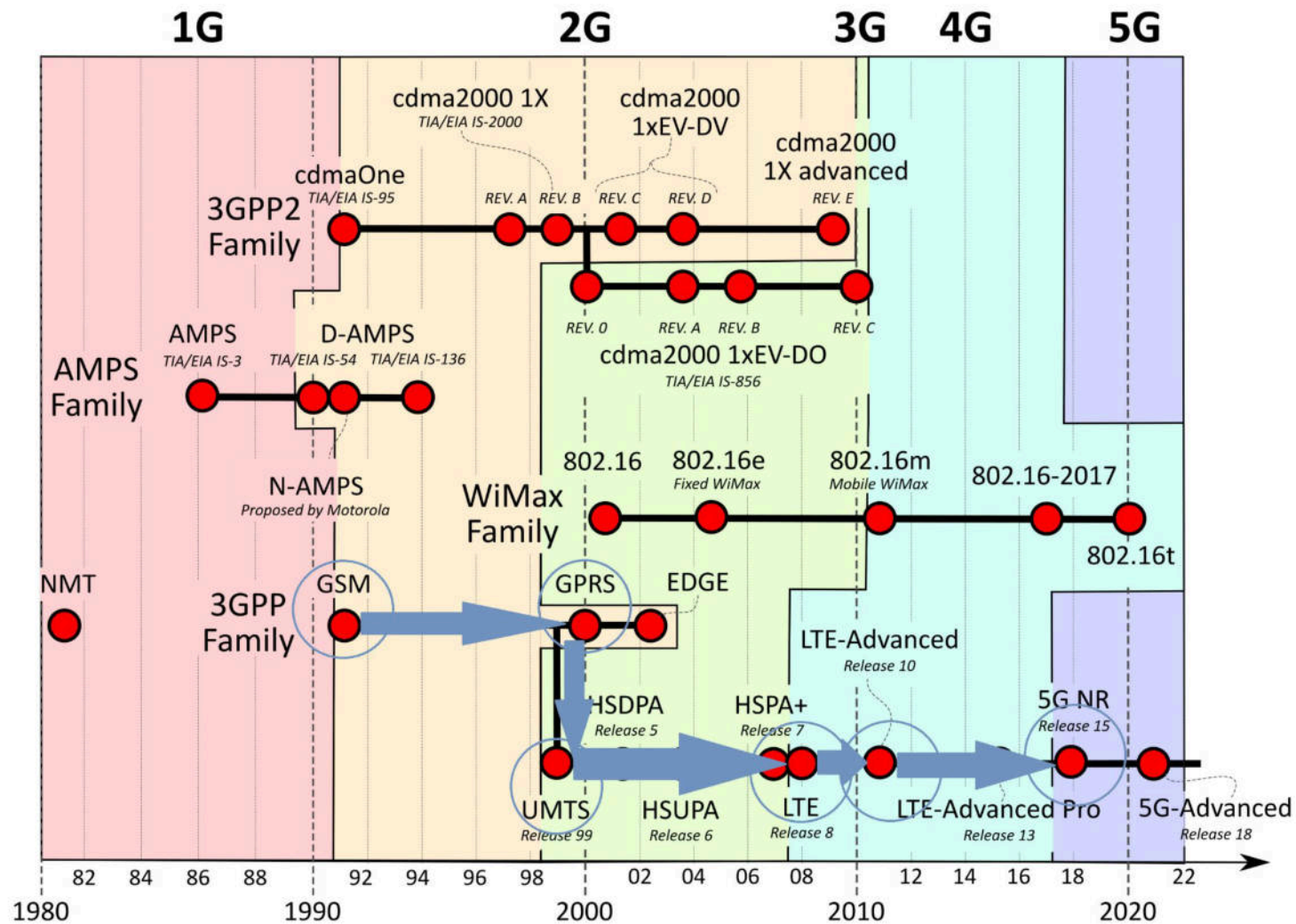
- Vulnerability Researcher at Trenchant
- RE and embedded dev
- Telco focused
- ~10 years of experience

What to cover

CALL ME



- 2G: GSM. Digital Technology! Voice and Data. Circuit Switched.
- GPRS: Both Circuit and Packet Switched!
- 3g: UMTS.
- LTE: Evolved packet core!
- 4G: LTE Advanced. VoLTE, IPv6, IMS.
- 5G: NR. Virtualization. Network slicing. Functionality generally split into "functions" instead of devices.

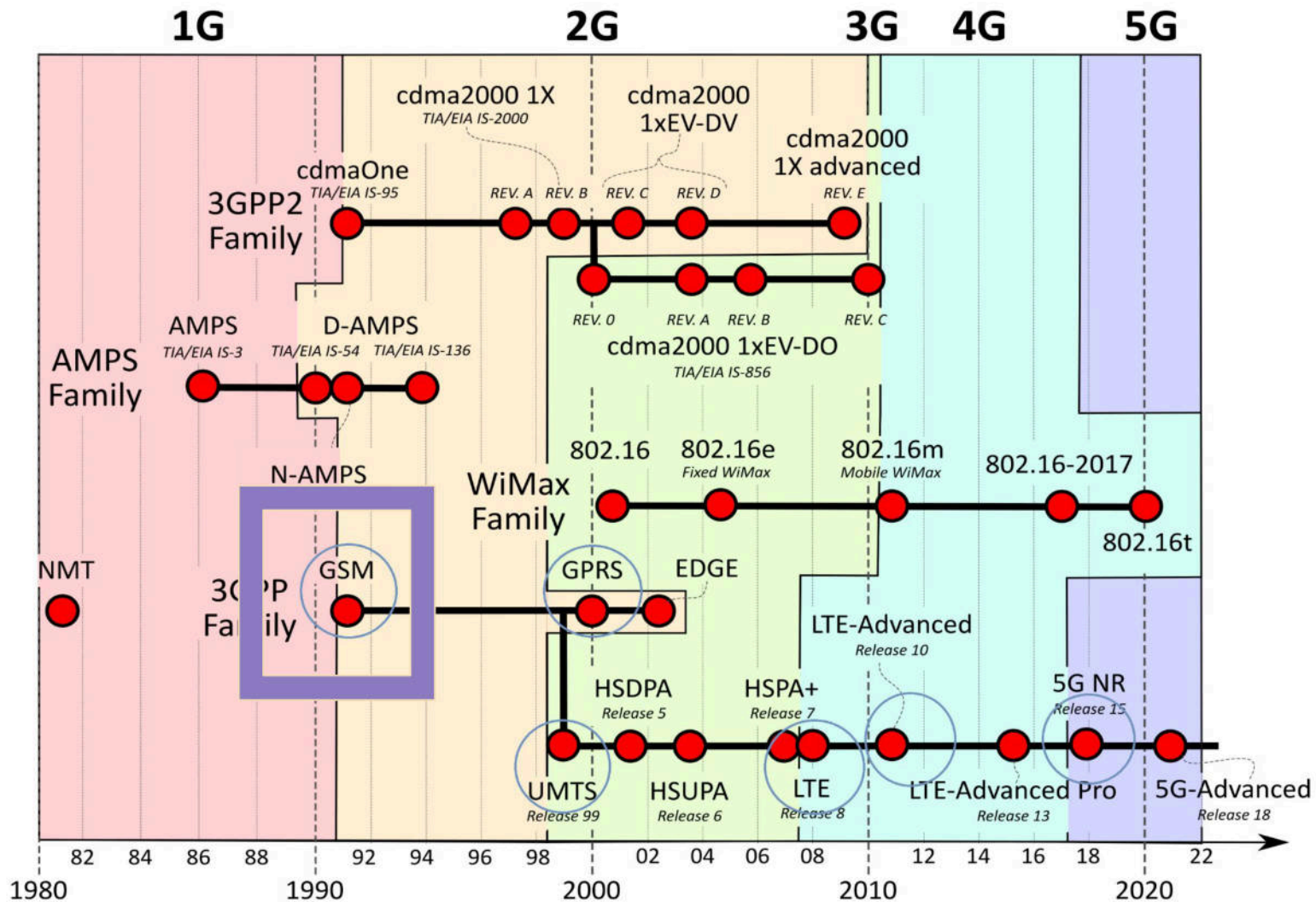


Basic Cellular Concepts

- **PSTN**
 - Public Switched Telephone Network
- Circuit Switched
- **PDN**
 - Public Data Network



Rachel Rosenbaum stars as Ridge in Amazon Studios' 'The Nervous Mrs. Miller'. (Amazon Studios)



2G

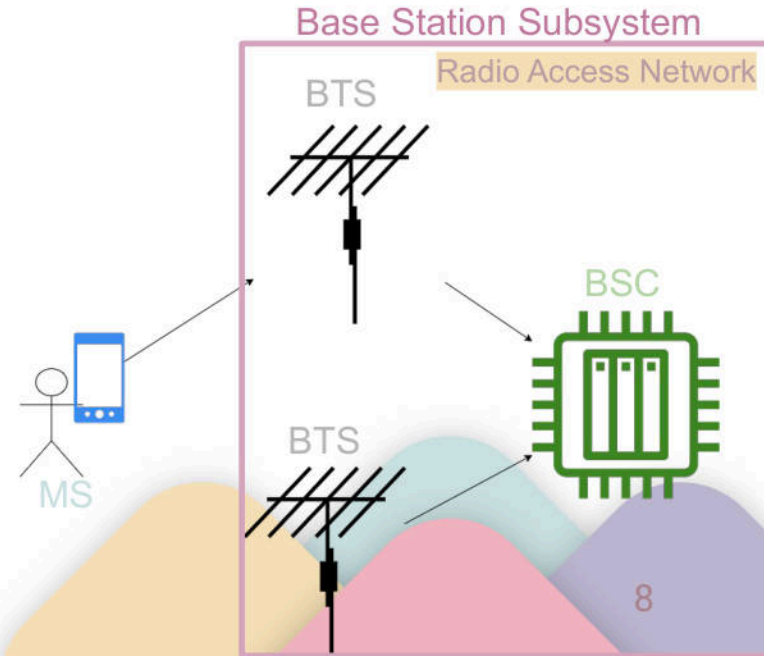
GSM



- Calls
- SMS
- ETSI and GSM standardizing!
- SS7 for signaling

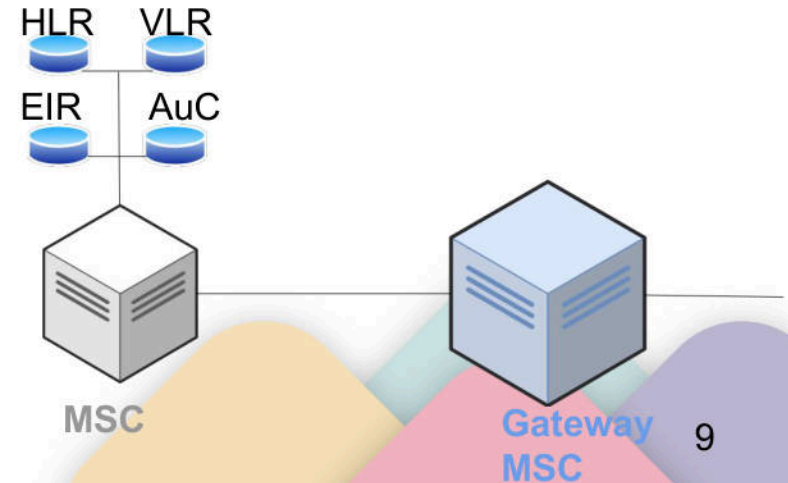
2G - GSM - Network Architecture

- Mobile Station (MS)
 - Device in your hand when you're making a call.
- Radio Access Network (RAN)
 - The Base Station Subsystem (BSS)
 - Base Transceiver Stations (BTS)
 - Base Station Controller (BSC)

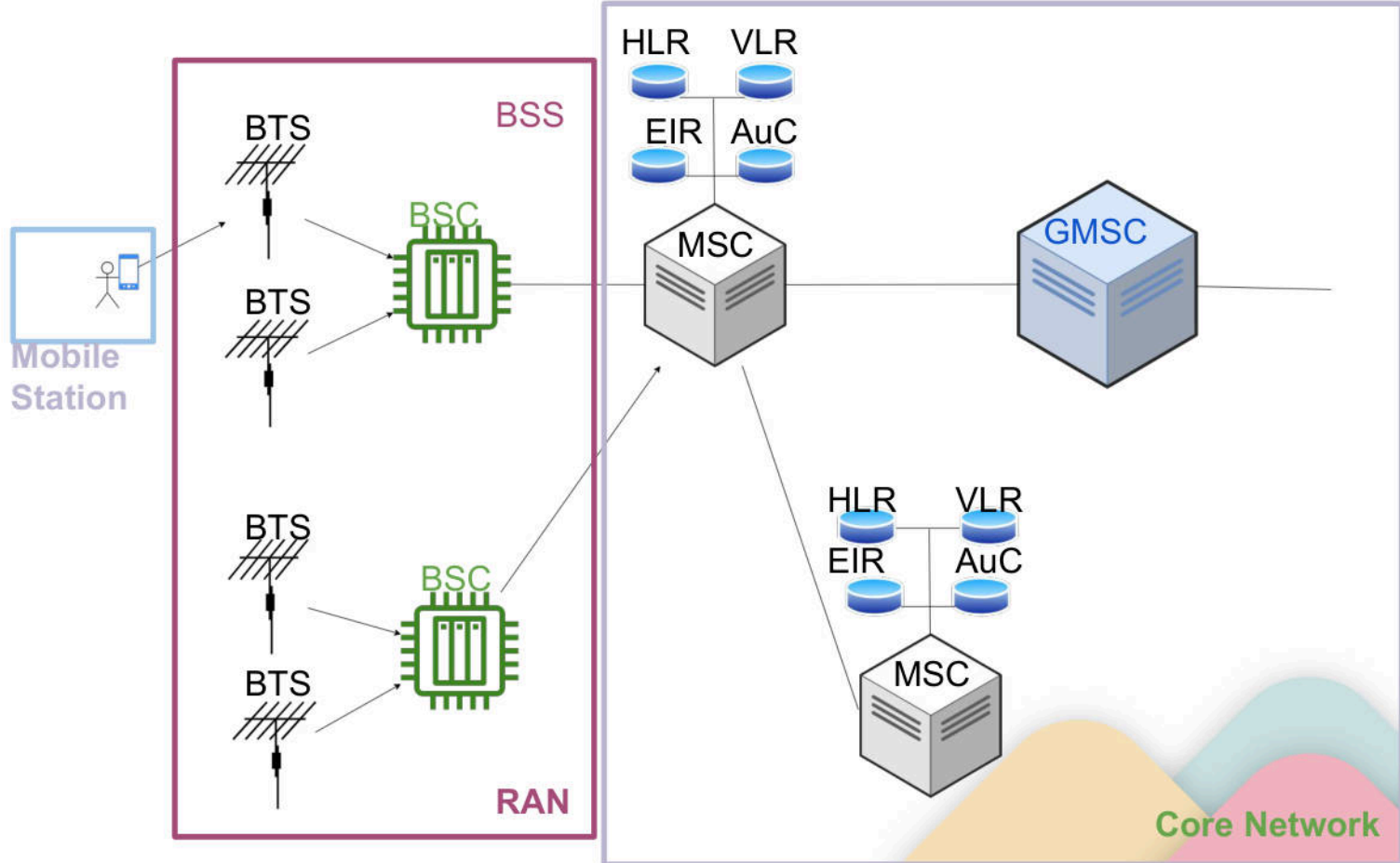


2G - GSM - Network Architecture

- Mobile Switching Center (MSC)
 - Handles the call switching, handover, registration, authentication.
- Home Location Register (HLR)
 - Large DB of user information for “local” users
- Visitor Location Register (VLR)
 - Large DB of user info for visiting users
- Authentication Center (AuC)
 - Secret keys stored here
- Equipment Identity Register (EIR)
 - Large DB of valid equipment on the network.



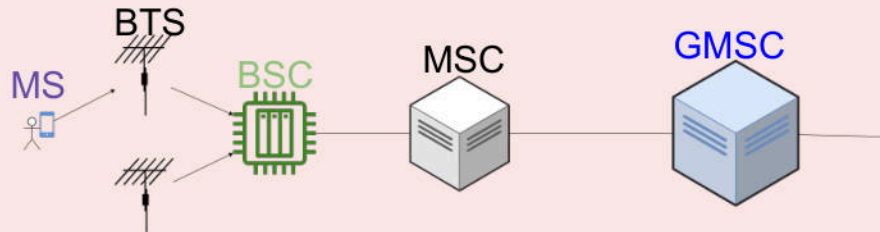
2G - GSM

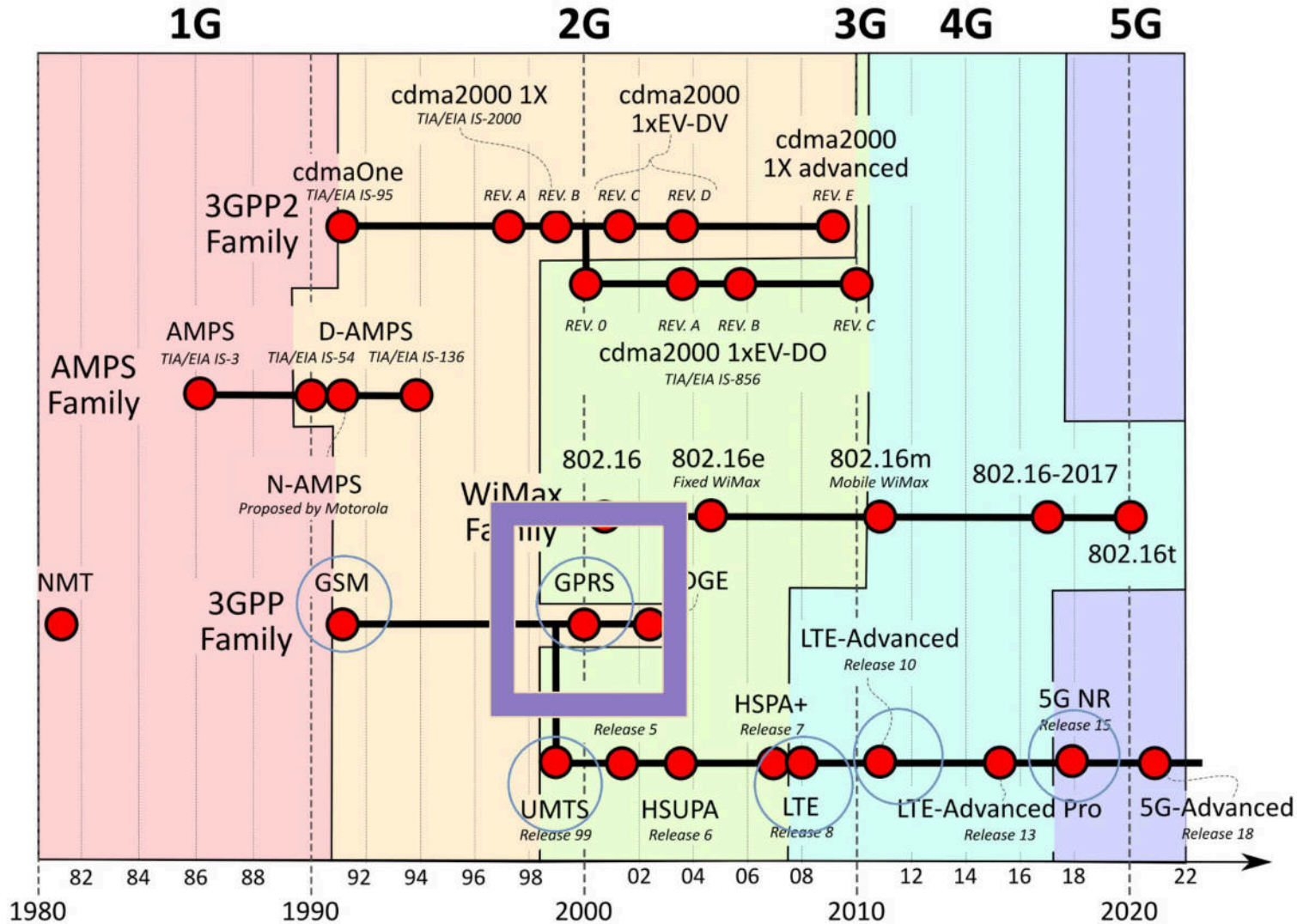


2G - GSM - Attack Vectors



- No mutual Authentication
 - IMSI-catchers / Stingray attacks
 - Rogue BTS
 - BTS determines encryption scheme!
- A5/1 is easily broken. A5/2 slightly harder but still weak.
- Jamming
- Clear text tx after BTS





2.5G

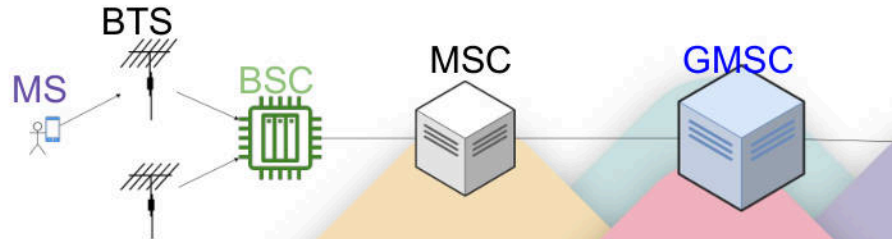
GPRS



- Packet Switched + Circuit Switched
 - GSM + IP
- More Services!
 - MMS
 - Push-To-Talk over cellular
 - 4+1 or 3+2 timeslots

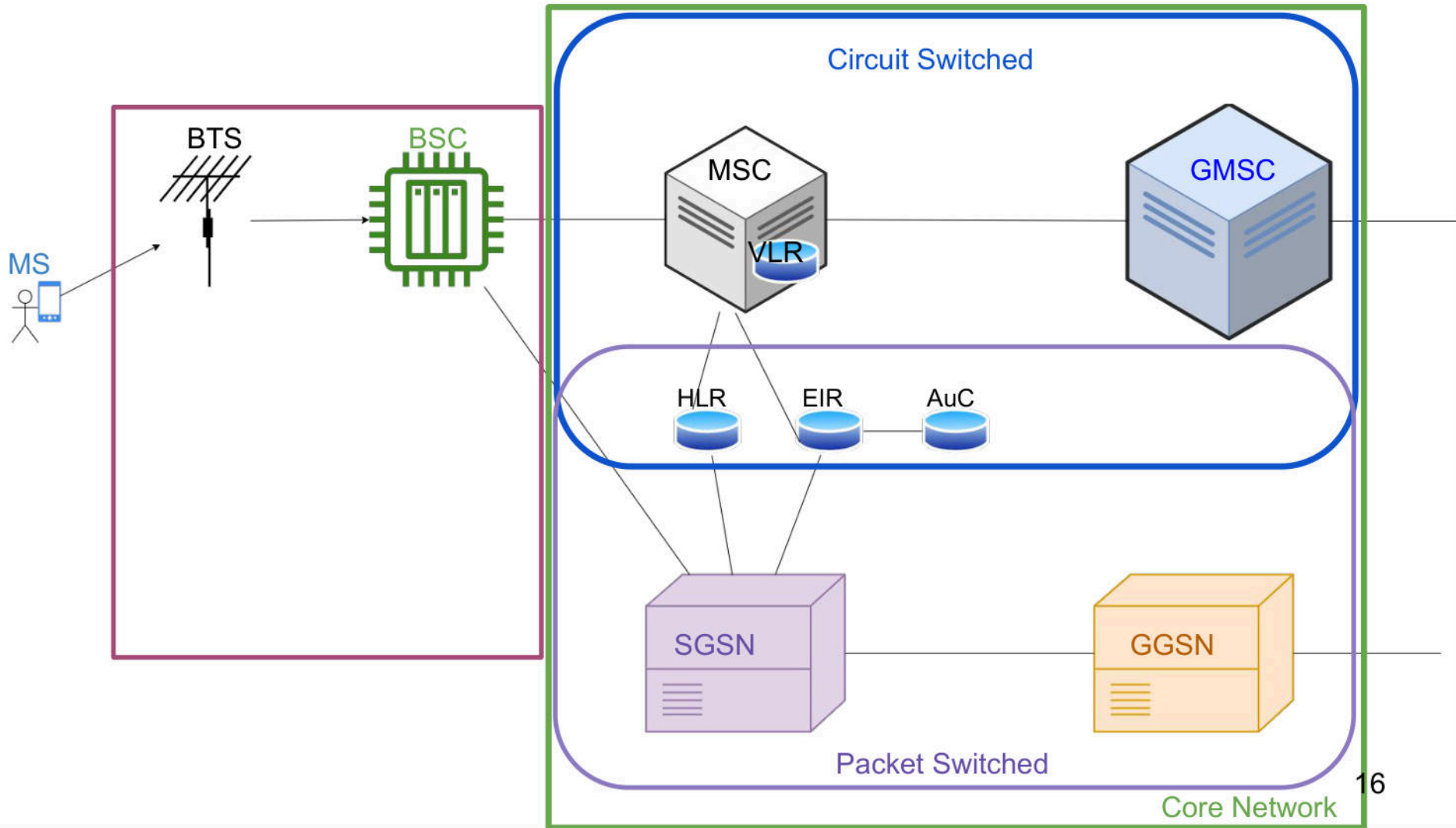
2.5G/GPRS - Network Architecture

- Front part mostly the same
 - Mobile Station (MS) / User Endpoint (UE)
 - Same thing.
 - BSS remains the same
 - BTS and BSC still there
- MSC functionality of before is needed for both portions of the network
 - Still feeds into the Gateway MSC



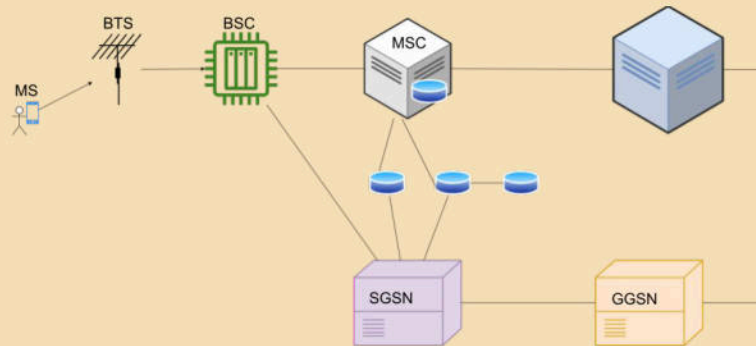
2.5G/GPRS - Network Architecture

- GPRS Support Nodes
 - Serving GPRS Support Node (SGSN)
 - location, sec and access controls, mobility management and charging
 - communicates with other **SGSN** in same network
 - reaches out to the **GGSN**.
 - GPRS Gateway Support Note (GGSN)
 - a router in the subnetwork.
 - Converts incoming IP traffic into GSM and hands that off to the **SGSN**.



Mitigations from 2G

- Mutual Authentication for Packet Switched Portion
- Authentication between Mobile Station and SGSN
- Some level of Ciphering algorithm for SGSN.
 - Some user data and control data still transmitted in plain text.
- SGSN identity verification.



2.5G/GPRS Attack Vectors

- Same known rogue BTS flaws
- Some implementations without any encryption
 - Italy and Denmark according to P1
- Nokia GGSN attack
 - non-standard IP options in a packet over the Gi interface
- Cisco GGSN ASR 5000
 - Improper Wireless Session Protocol packet handling
 - Portal Page bypass
 - Improper HTTP packet handling
 - Send requests through GGSN without Verifying!

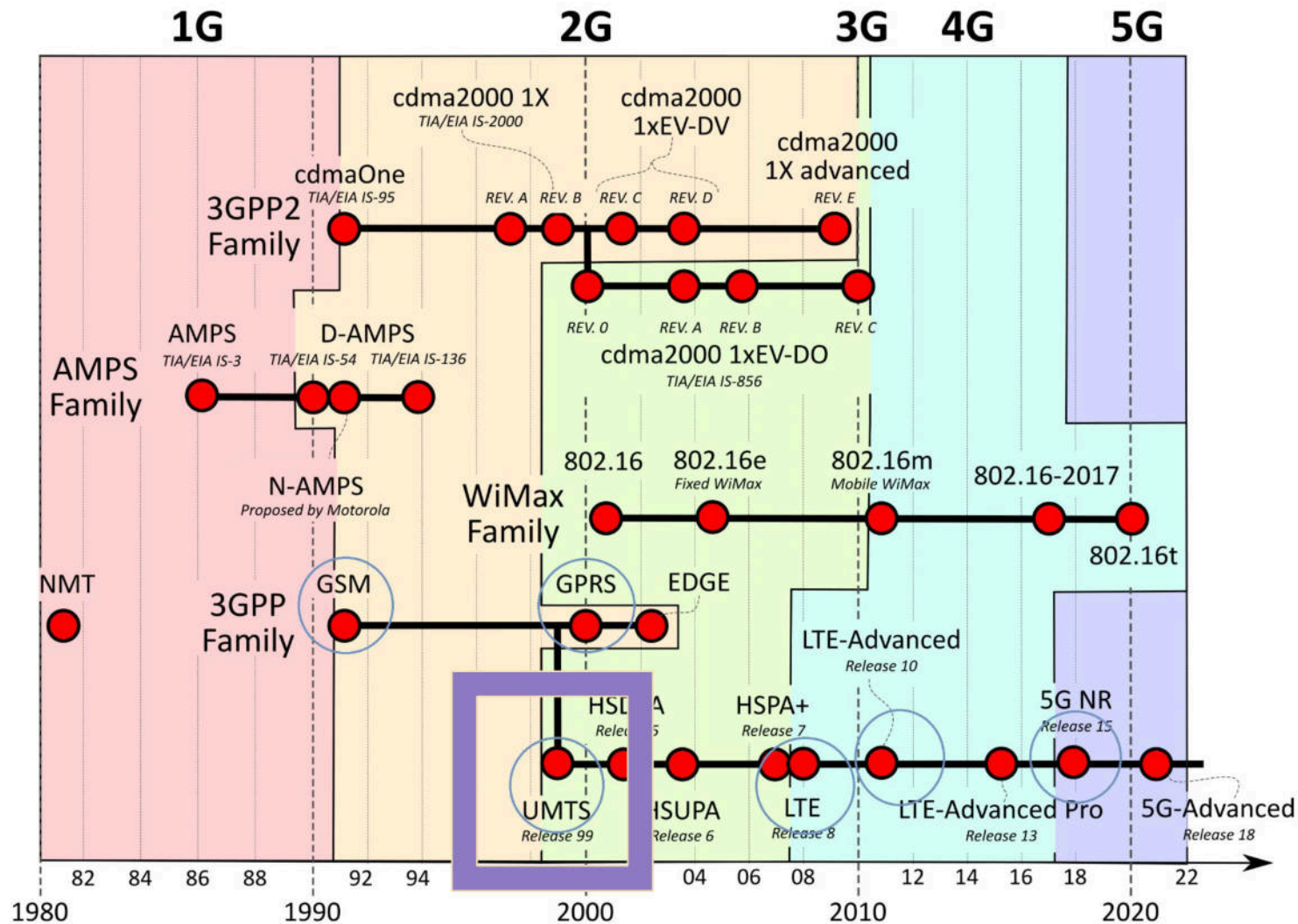


3G

UMTS



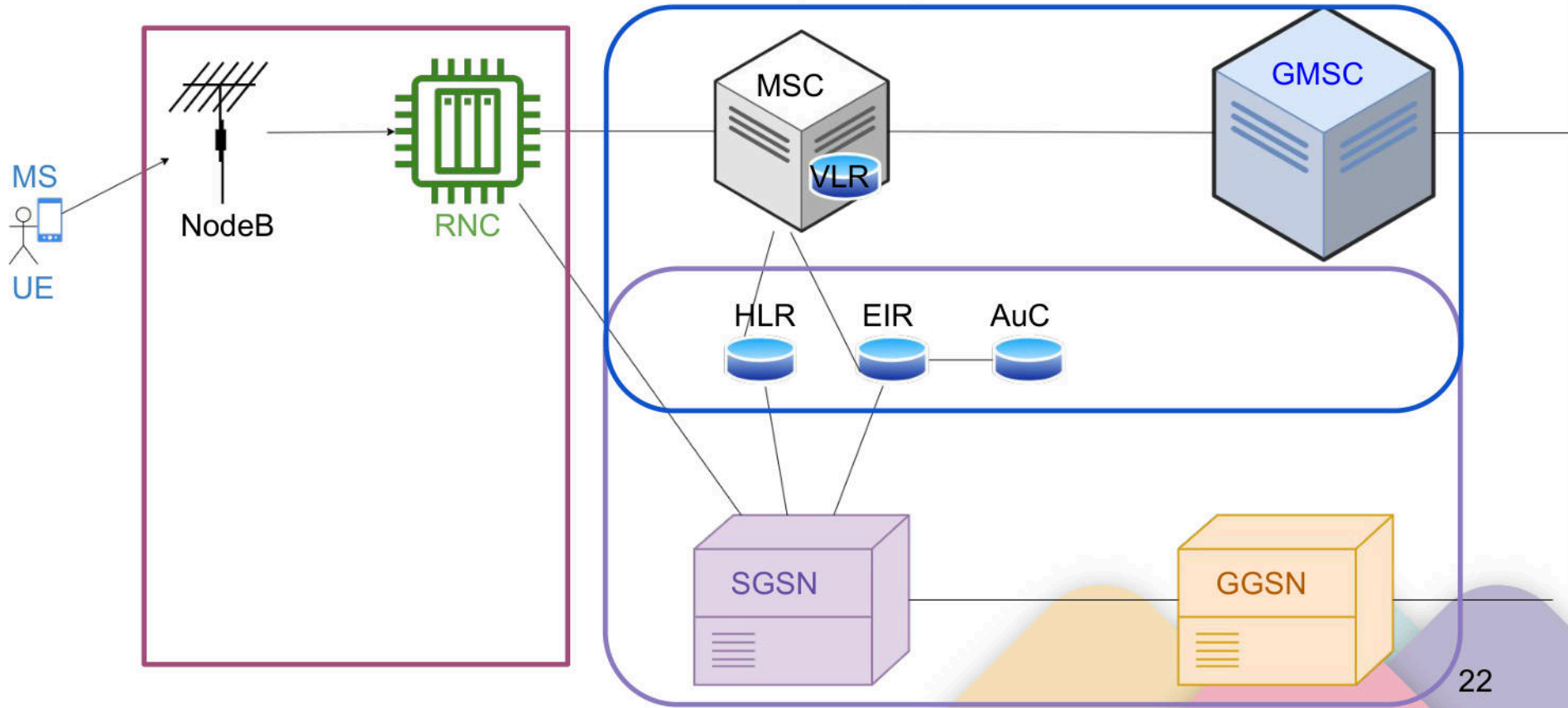
- More internet functionality!
- Video Chat!
- Mobile Broadband!
- SIGTRAN



3G - UMTS -Network Architecture

- Kept most of 2.5G
- RAN is now called UTRAN (**UMTS terrestrial Radio Access Network**)
- NodeB
 - This is literally the same basic placement in the network and functionality as the
- Radio Network Controller (RNC)
 - In charge of all NodeBs for radio management.
 - Some Mobility Management
 - User data encryption

3G - UMTS - Architecture



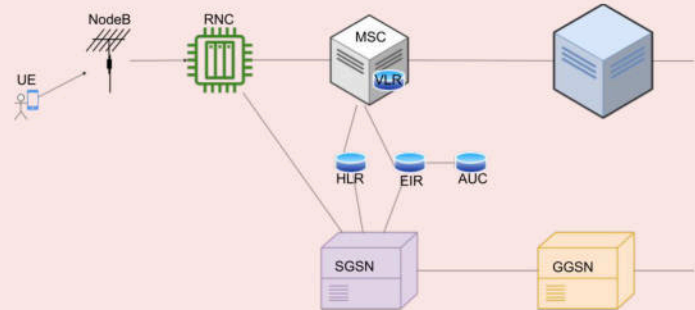
Mitigations from 2.5G

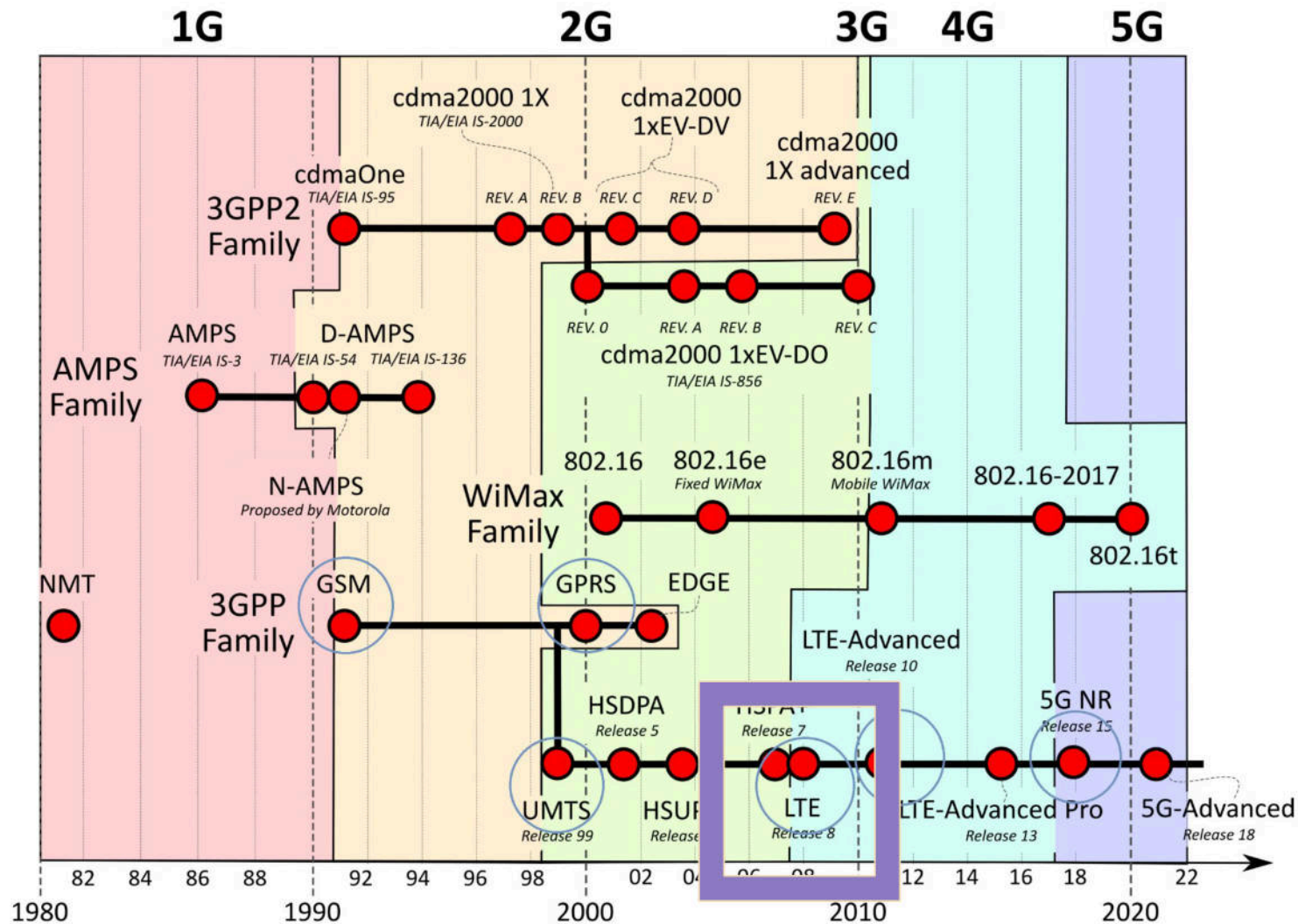
- True Mutual Auth!
- (Attempts at) IMSI catcher mitigation
- Radio Access Link preserves
 - IMSI confidentiality
 - User Location confidentiality
- Signaling plane and user plane now have confidentiality requirements in the specs!
- Security should now be shown to the user
 - eg when hand off occurs to 2G



3G - UMTS - Attack Vectors

- Rogue **NodeB** (BTS) can still happen
- Downgrade Attacks
- Remote IMSI Attack
 - Integrity keys between **UE** and **RNC** are generated in the Core Network and tx unencrypted to RNC, and sometimes between RNCs
- **HLR** overloading using SQN
 - **rrcConnectionRequest** messages
- Cisco ASR 5000 Series GGSN
 - Not properly handling TCP packets





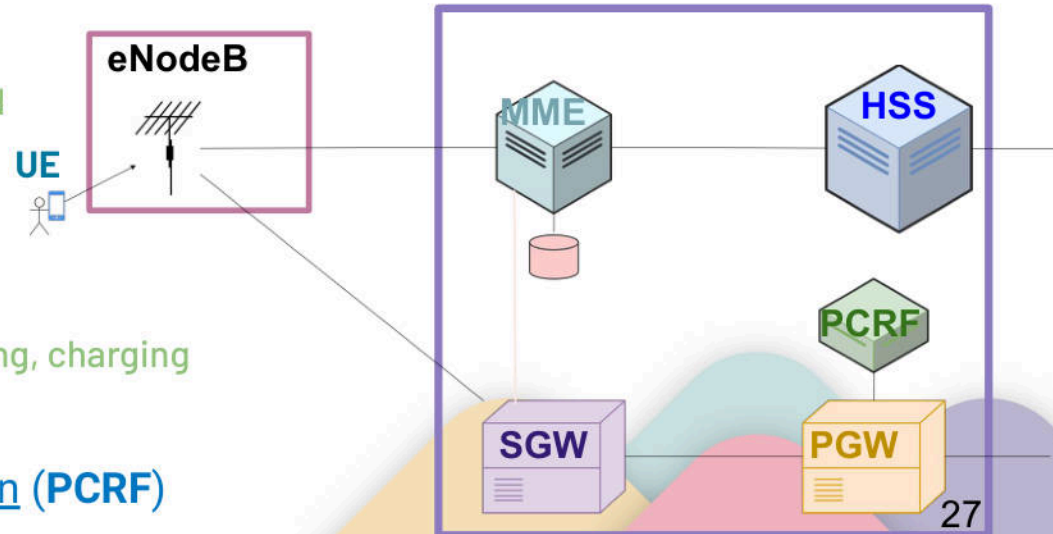
LTE



- 3.95G? 4G? Guess it's called 4G now?
- Core network is now Evolved Packet Core aka System Architecture Evolution
- VoLTE, carrier adopted improvements
- Diameter

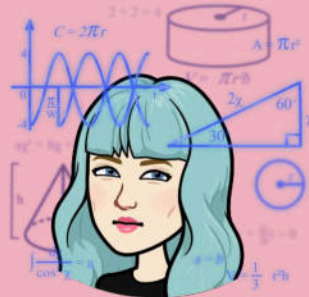
LTE - Network Architecture

- eNodeB
- Mobility Management Entity (MME)
 - replaces the RNC
 - Directs to appropriate SGW.
- Home Subscriber Server (HSS)
 - HLR and Authentication combined
- Signal Gateway (SGW)
 - Routes and Forwards packets.
- PDN Gateway (PGW)
 - policy enforcement, packet filtering, charging
 - Gateway to the outside
- Policy and Charging Rules Function (PCRF)



Mitigations

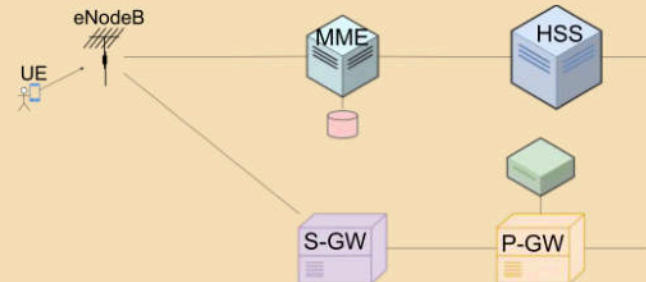
- eNodeB security?!
- IMEI is not sent to eNodeB until security established to Core Network
- Encryption Algorithm based on AES
 - Null X algorithm (Null ciphering algorithm, Null integrity protection algorithm)
- Mobility Management is now split apart and in its own entity

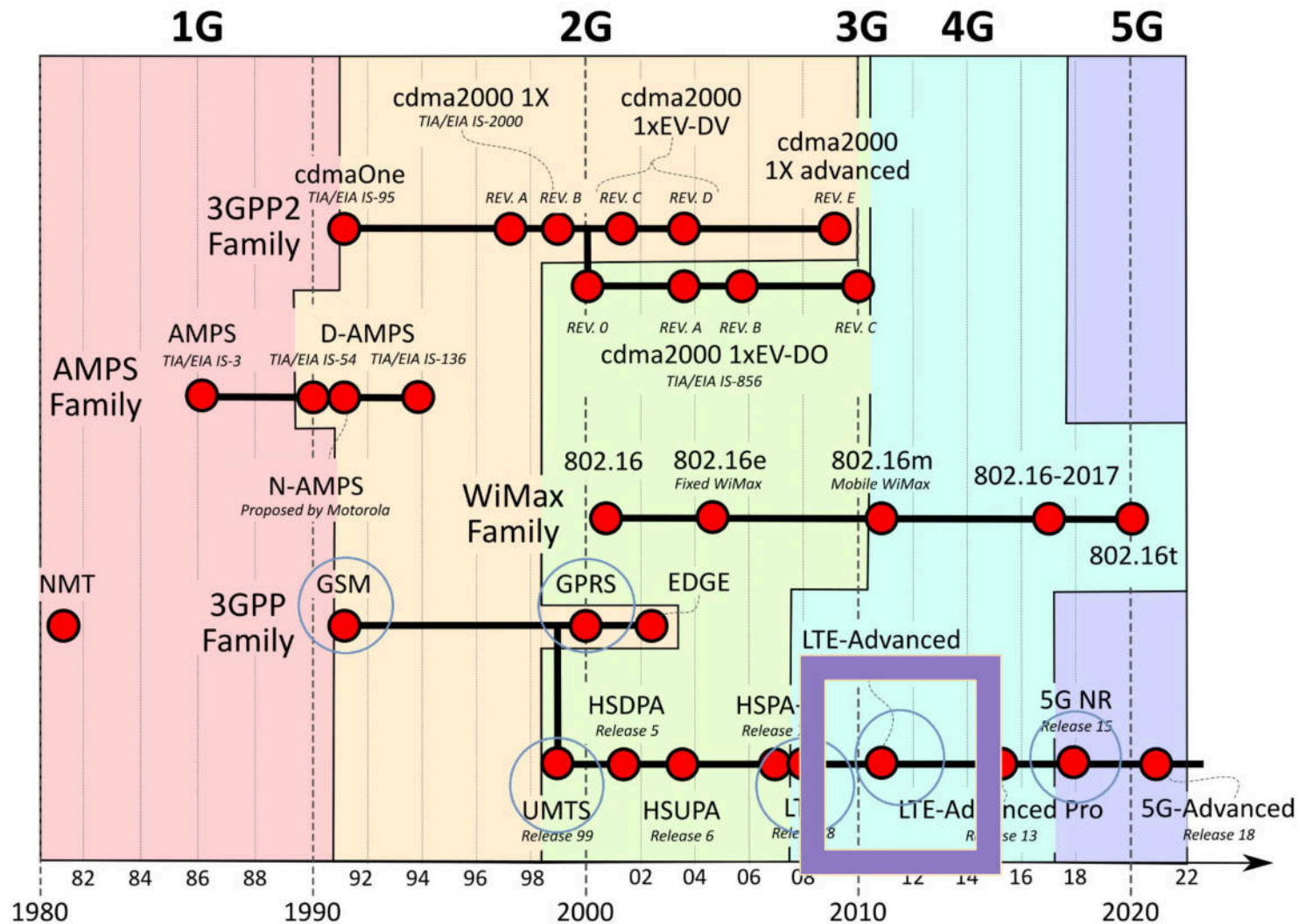


LTE - Attack Vectors



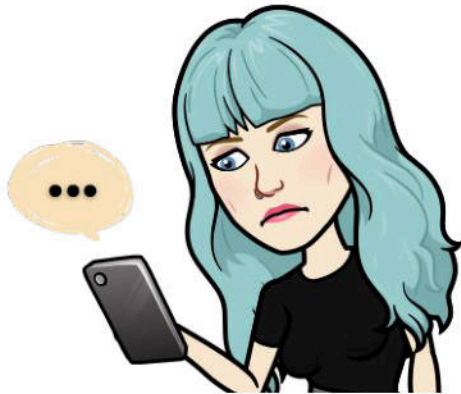
- LTEInspector
 - Authentication Relay Attacks
 - Paging Channel Hijacking Attack
 - Panic Attack
- Actual Implementation Errors for Identifiers allowing location tracking
 - Temporary Identifier not being temporary
- Cisco PGW 2200 Remote DOS





4G

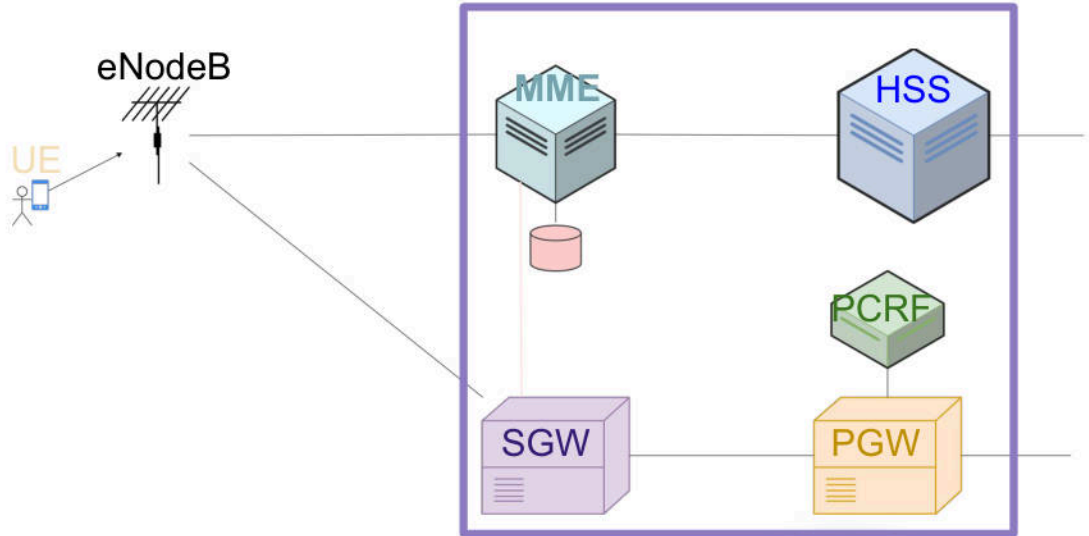
LTE Advanced, LTE Advanced Pro



- IPv6 expansions
- Adaptive modulation
- Time varying channel
- Voice over LTE
- Carrier Aggregation (LTE Advanced) and enhanced carrier aggregation (LTE Advanced Pro)
- IP Multimedia Subsystem

4G - LTE Adv - Network Architecture

- Same network architecture as 3.95G
- UE
- eNodeB
- MME
- HSS
- SGW
- PGW
- PCRF



Mitigations from LTE and Prior

- All radio interface data shall be encrypted
- AKA procedure for mutual auth between **User Endpoint** and **Evolved Packet Core**
- Different session keys for encryption and integrity protection
- Mobile Network Operator may provide confidentiality in Radio Resource Control signaling.
- Secondary authentication (for data network outside Mobile Network Operator)



4G - LTE Adv - Attack Vectors

- Still have IMSI catchers!
- Cisco ASR 5000 PGW DoS
 - Vector Packet Processor Error
 - GTP Header processing error
- MME vuln
 - P1 HITB talk on keys being hardcoded
- MML Interface exposed to the wide internet on some routers





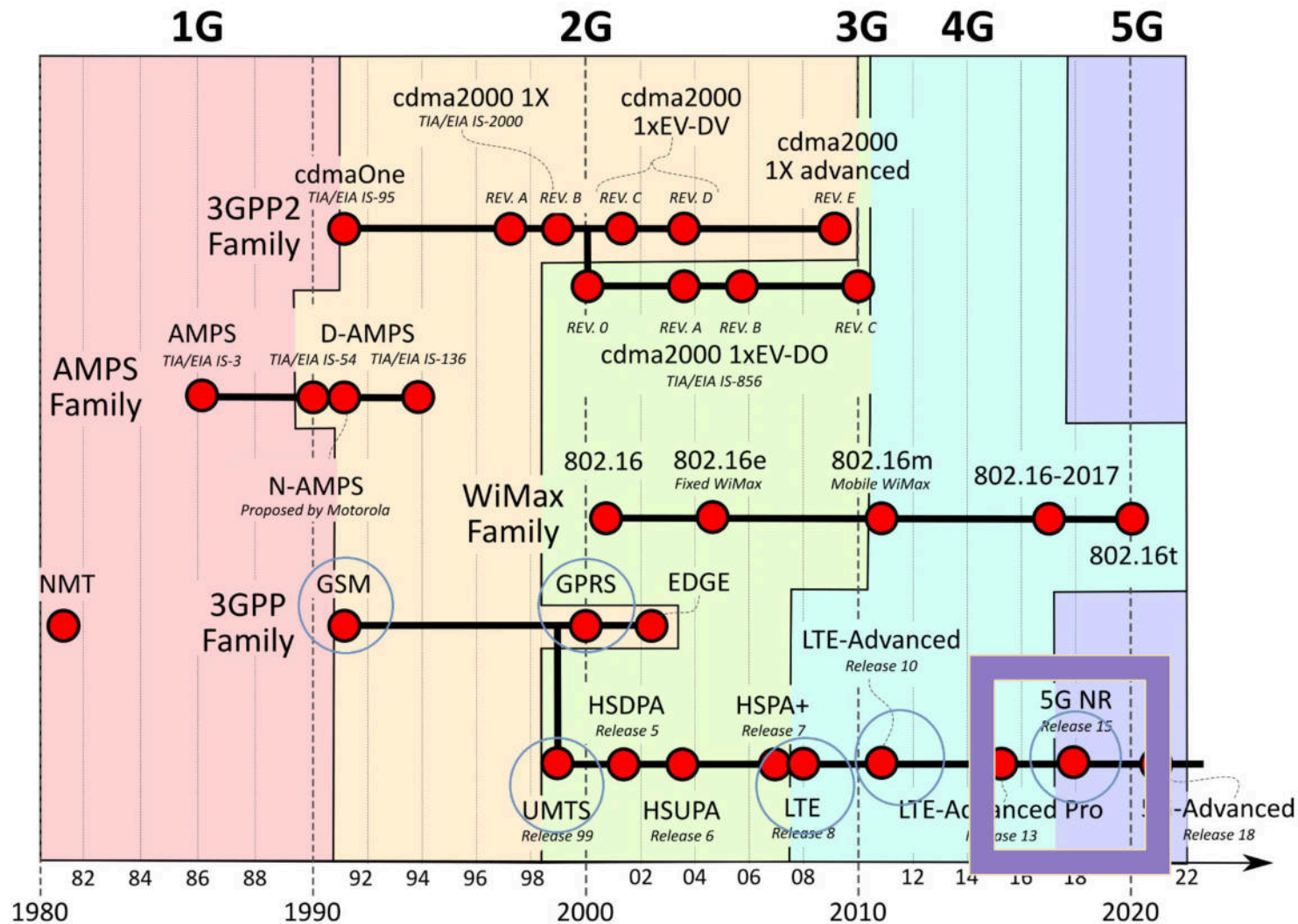
5G



Not just for telephony!



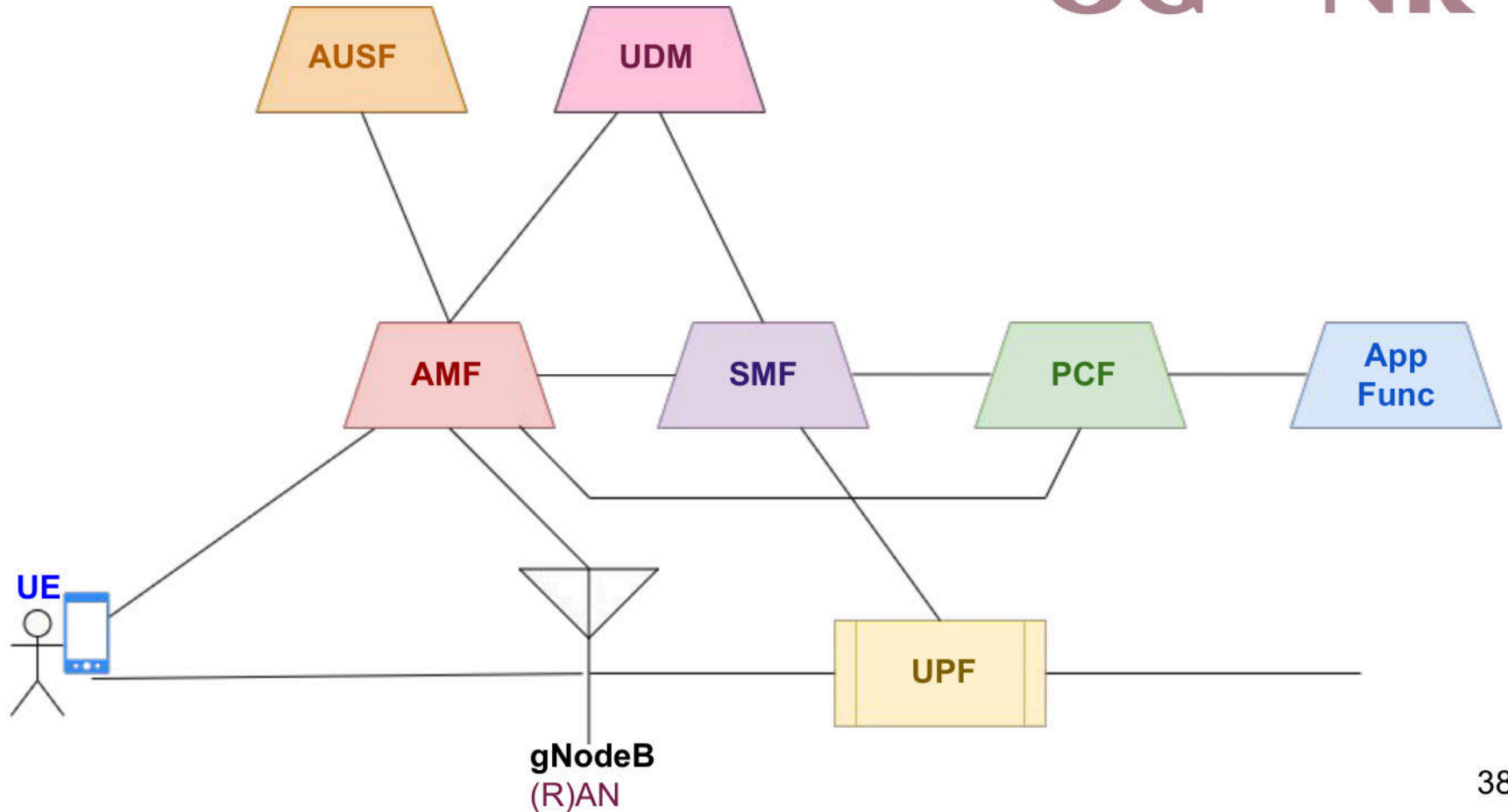
- Network Function Virtualization
- Management and Orchestration is now officially part of the core network
- New concept of Network Slicing
 - Different devices have different needs!
- Renaming the eNodeB to **gNodeB** just cuz!



5G - NR - Network Architecture

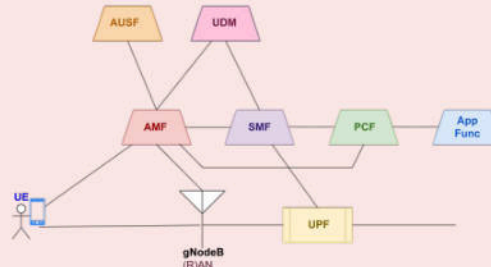
- Access and Mobility Management Function (AMF)
 - Mobility Management, connection management, registration management
- Session Management Function (SMF)
 - Session management, data network profile. UPF selection. IP address allocation if IP based
- User Plane Function (UPF)
 - Anchor for RAN mobility and QoS flow, Policy enforcement
- Unified Data Management (UDM)
 - Access authorization, Data network profiles. Registration and Mobility Management
- Policy Control Function (PCF)

5G - NR



Mitigations from 4G

- Subscription Concealed Identifier (SUCI) and Subscription Permanent Identifier (SUPI) to protect IMSI
- Security and Mobility separated in the core network
- 256bit keys supported
- Perimeter security to protect user plane messages (GTP-U).
- **Authentication Server** and **UE** add a counter to mitigate replay attacks.
- Security Anchor Function (SEAF) allows AKA to skip full reauth



5G Attack Vectors

- RA and CN don't require authentication.

Location sniffing attack

- Place sniffing tool near target UE
 - Monitor for plaintext **AuthReq** msg
 - The message contains **RAND** and **AUTN**
- Build an **AuthReq** message using that **RAND** and **AUTN**.
- Broadcast fake message.
 - MAC check fails if not target UE
 - Victim UE passes MAC, fails SEQ
 - If all messages fail MAC verification, the UE is not nearby. If one SEQ failure, it is nearby.

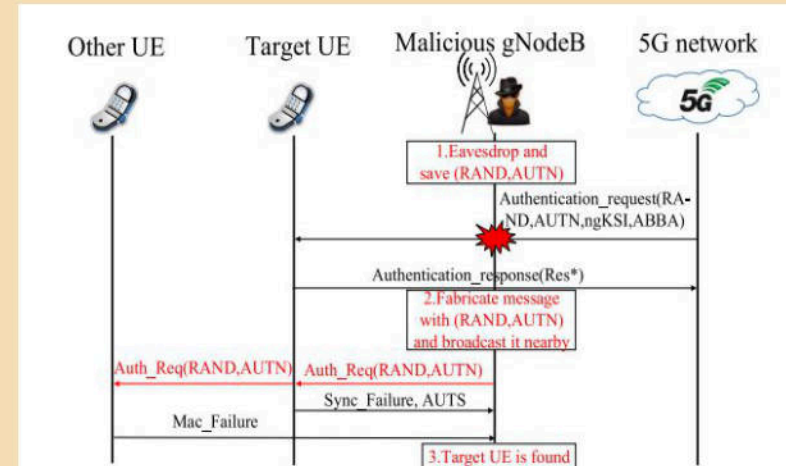


Fig. 2 Location sniffing attack.

5G Attack Vectors

- Attach Req Injection
- Battery Draining
 - NB-IOT and LTE-M devices in particular can be drained like 5x faster than expected
- Sending malformed packets can trigger crashes in 5g cores. Yes, still.
- HTTP/2 Attacks
 - Stream Reuse
 - Closed Stream
- End-to-end China Unicom



Review

**Sorry, I wasn't
listening.**



- **2G:** Digital Technology! Voice and Data. Circuit Switched. User, BSS, MSC, GMSC.
- **GPRS:** Both Circuit and Packet Switched! Added the SGSN and GGSN.
- **3G:** UMTS NodeB, RNC, kept SGSN and GGSN.
- **LTE:** Evolved packet core! eNodeB, Mobility Management, Signal Gateway and PDN Gateway.
- **4G:** VoLTE, IPv6, IMS. Kept devices same.
- **5G:** Virtualization. Network slicing. gNodeB. Functionality generally split into "functions" instead of devices.

References

- Most used image came from https://en.wikipedia.org/wiki/Cellular_network
- The Critical Hole at the Heart of Our Cell Phone Networks, Zetter
<https://www.wired.com/2016/04/the-critical-hole-at-the-heart-of-cell-phone-infrastructure/>
- Taking up the Gauntlet - SS7 Attacks in Ukraine, Cathal Mc Daid, <https://blog.adaptivemobile.com/russia-ukraine-telecom-monitoring>
- <https://www.cnvd.org.cn/flaw/show/CNVD-2014-00546> Cisco ASR 5000 Series Devices GPRS Support Node Security Bypass Vulnerability
- <https://www.cisco.com/c/en/us/support/docs/csa/cisco-sa-cisco-pdng-dos-KmzwEy2Q.html> Cisco Packet Data Network Gateway IPsec ICMP Denial of Service Vulnerability
- <https://sec.cloudapps.cisco.com/security/center/content/CiscoSecurityAdvisory/cisco-sa-20160621-asr> Cisco ASR 5000 Series Packet Data Network Gateway Denial of Service Vulnerability
- [LTEInspector: A Systematic Approach for Adversarial Testing of 4G LTE](#), Syed Rafiul Hussain, Omar Chowdhury, Shagufta Mehnaz, Elisa Bertino
- Nokia warns 5G security 'breaches are the rule, not the exception', Kapko, <https://www.cybersecuritydive.com/news/5g-security-breaches/636693/>
- 5G Network Attacks Projects, <https://networksimulationtools.com/5g-network-attacks-projects>
- P1 security Archives, <https://www.p1sec.com/corp/category/p1-security/>
- 3G Security; Security Threats and Requirements (Release 4), 3GPP TS 21.133 V4.0.0, 3rd Generation Partnership Project, 2003; www.3gpp.org/ftp/tsg_sa/wg3_security/_specs/Old_Vsns/21133-400.pdf
- [A Vulnerability in 5G Authentication Protocols and Its Countermeasure \(2020\)](#). Xinxin HU, Caixia LIU, Shuxin LIU, Jinsong LI, and Xiaotao CHENGHU. IEICE Transactions on Information and Systems. E103.D. 1806-1809. 10.1587/transinf.2019FOL0001.
- Privacy Attacks to the 4G and 5G Cellular Paging Protocols Using Side Channel Information. Syed Rafiul Hussain, Mitziu Echeverria, Omar Chowdhury, Ninghui Li and Elisa Bertino.
<https://homepage.divms.uiowa.edu/~comarhaider/publications/LTE-torpedo-NDSS19.pdf>

References

- On the Detection of Signaling DoS Attacks on 3G WiMax Wireless Networks. Patrick P.C. Lee, Tian Bu , Thomas Woo
<https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=cb5197dfceb76fe6ac37909c6ef935285f9e0cc7>
- Statement of Vulnerabilities in ZTE MF910 and MF65+ Products. ZTE. 2019.
<https://support.zte.com.cn/support/news/LoopholeInfoDetail.aspx?newsId=1010203>
- ZTE MF910 – An end of life router, running lots of vivacious hidden code. Richter, Pentest Partners.
<https://www.pentestpartners.com/security-blog/zte-mf910-an-end-of-life-router-running-lots-of-vivacious-hidden-code>
- LTE Pwnage: Hacking HLR/HSS and MME Core Network Elements P1 Security. Presented at HITBSecConf2013. Amsterdam.
<https://conference.hitb.org/hitbsecconf2013ams/materials/DIT2%20-%20Philippe%20Langlois%20-%20Hacking%20HLR%20HSS%20and%20MME%20Core%20Network%20Elements.pdf>
- Privacy Attacks to the 4G and 5G Cellular Paging Protocols Using Side Channel Information. Syed Rafiul Hussain, Mitziu Echeverria, Omar Chowdhury, Ninghui L., Elisa Bertino.
- New 4G LTE Network Attacks Let Hackers Spy, Track, Spoof and Spam. Swati Khandelwal. The Hacker News, Mar 05, 2018.
<https://thehackernews.com/2018/03/4g-lte-network-hacking.html>
- GUTI Reallocation Demystified: Cellular Location Tracking with Changing Temporary Identifier. Byeongdo Hong, Sangwook Bae, Yongdae Kim. Presented at the Network and Distributed System Security Symposium.
https://syssec.kaist.ac.kr/pub/2018/hong_ndss_2018.pdf
- A first look on the effects and mitigation of VoIP SPIT flooding in 4G mobile networks. Bou-Harb, Elias & Debbabi, Mourad & Assi, Chadi. (2012). 982-987. 10.1109/ICC.2012.6364233.
- Cheating VoIP Security by Flooding the SIP. Irfan Shakeel. Jan 20, 2016.
<https://resources.infosecinstitute.com/topics/hacking/cheating-voip-security-by-flooding-the-sip/>

More References!

- New Vulnerabilities in 5G Networks. Altaf Shaik*, Ravishankar Borgaonkar. Presented at Black Hat Briefings. <https://i.blackhat.com/USA-19/Wednesday/us-19-Shaik-New-Vulnerabilities-In-5G-Networks-wp.pdf>
- Cisco PGW 2200软件交换机多个安全漏洞 May 2010. CNVD-2010-0844. <https://www.cnvd.org.cn/flaw/show/CNVD-2010-0844>
- LTE核心网存在拒绝服务漏洞. Jan 2021. CNVD-2020-71678. <https://www.cnvd.org.cn/flaw/show/CNVD-2020-71678>
- Cisco Gateway GPRS Support Node TCP Invalid Packet Vulnerability. Cisco. CVE-2015-4201. <https://sec.cloudapps.cisco.com/security/center/content/CiscoSecurityAdvisory/Cisco-SA-20150619-CVE-2015-4201>
- <https://www.mpirical.com/> for 5G trainings
- Signalling Security Analysis: Is HTTP/2 Secure in 5G Core Network? Xinxin Hu, Caixia Liu, Shuxin Liu, Wei You, Yu Zhao; 2018 10th International Conference on Wireless Communications and Signal Processing. Hangzhou, China, 2018. doi: 10.1109/WCSP.2018.8555612.
- Bookworm Game: Automatic Discovery of LTE Vulnerabilities Through Documentation Analysis; Yi Chen, Yepeng Yao, XiaoFeng Wang, Dandan Xu, Chang Yeue, Xiaozhong Lui, Kai Chen, Haixu Tang, Bauxu Liu. 2021 IEEE Symposium on Security and Privacy IEEE publication
- [ESF Potential Threats to 5G Network Slicing](#), NSA, CISA. <https://www.cisa.gov/news-events/alerts/2023/07/17/nsa-cisa-release-guidance-security-considerations-5g-netw>
- [New Vulnerabilities in 5G Networks-Blackhat Briefings](#) Altaf Shaik*, Ravishankar Borgaonkar. <https://i.blackhat.com/USA-19/Wednesday/us-19-Shaik-New-Vulnerabilities-In-5G-Networks-wp.pdf#fork-slicing>
- Statement of Vulnerabilities in ZTE MF910 and MF65+ Products. ZTE. 20 February 2019. <https://support.zte.com.cn/support/news/LoopholeInfoDetail.aspx?newsId=1010203>
- Potential Threat Vectors to 5G Infrastructure – U.S. Department of Defense. May 2021. <https://media.defense.gov/2021/May/10/2002637751/-1/-1/1/POTENTIAL%20THREAT%20VECTORS%20TO%205G%20INFRASTRUCTURE.PDF>



Questions?



Thank you!!!

- My cow
- Feedback and

